Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms.

Monicity check C002: a mathematical proof and its later role. The intended Γ-morphisms have bijections on full RV preimages. For these categories, the horizontal positive-semiring maps are indeed injective by the following argument. It uses a birational classification theorem of Reichstein–Youssin, not an HK input. That theorem is not currently an accepted Lean input and has not been proved locally; thus this is a mathematical proof for review, not a report that the general monomorphism has been formalized. No new premise has been added to Lean.

Let $F : I^\sharp \longrightarrow J^\sharp$ be an original-ground definable RV bijection in degree $n > 0$, retaining all finite disjoint-union tags. Each full value fibre is a torsor for $(\Bbbk^\times)^n$ and has a unique residue-generic type. Residue/Gamma orthogonality makes $\mathrm{vrv} \circ F$ finite-valued on that fibre. Exactly one target value and tag receives its generic part. The image has residue dimension n, hence is generic in its target fibre. Applying the same argument to F^{-1} gives an original-ground definable bijection $h : I \longrightarrow J$. Definability follows from definability of residue dimension in families. The original F need not contract on its exceptional subsets.

Fix $\alpha \in I$ and put $\beta = h(\alpha)$ and $\Delta = \mathrm{vrv}(\mathrm{RV} \cap \mathrm{dcl}(\mathbb{S}))$. Form the finitely generated abelian group

$$L = (\Delta + \textstyle\sum_i \mathbb{Z}\alpha_i + \sum_i \mathbb{Z}\beta_i)/\Delta.$$

The diagonalizable group D with character group L acts on the two residue tori with weights $\bar{\alpha}_i$, $\bar{\beta}_i$. Indeed every character $L \longrightarrow \Bbbk^\times$ extends to Γ/Δ; the resulting RV twist $u \mapsto \chi(\mathrm{vrv}(u))u$ fixes the old ground, the residue field and every Gamma value. Relative RV quantifier elimination makes it preserve F. In residue coordinates the generic restriction of F is consequently D-equivariantly birational. The actions have equal kernels by this birational map and its inverse; their joint kernel is trivial because their weights generate L. Both actions are therefore faithful.

By Reichstein–Youssin, *A birational invariant for algebraic group actions*, Theorem 7.1 and Corollary 2.5 (arXiv:math/0007181), there is $M \in \mathrm{GL}_n(\mathbb{Z})$ with $\beta - M\alpha \in \Delta^n$. Choose old-ground definable RV points c_i having these values. Then

$$H(u)_i = c_i \prod_j u_j^{M_{ij}}$$

bijects the full fibres at α and β. This is a replacement for the generic map, not an assertion that the given rational map extends everywhere. Compactness gives finitely many such formulas on a definable partition of I; they glue to a full RV bijection contracting to h. Thus h is a $\Gamma[n]$-isomorphism. Degree zero is the finite-tag case. This proves the lower positive injection; the upper follows from the commuting square and the two vertical positive injections. It does not assert injectivity after groupification.

Later development. Horizontal monicity is unnecessary for the balanced tensor map, the corrected Proposition 3.18, or the retractions. The latter need their balanced evaluations to respect the corrected kernel congruences. In the unmeasured case an elementary pair has the same value because $\chi(U) = \chi(U + \delta e_i)$, including when δ has no old-ground RV representative. In the measured case the translation transports the complete combined weight unchanged. The annular-ideal pullback uses its commuting square and vanishing under forgetting forms, not coefficient injectivity.

The later uses of Ψ^{-1} and the bounded comparison still need the quotient formulation required by the correction of Proposition 3.18. The piecewise retractions h_m have their own descent checks. Neither issue is resolved or worsened by horizontal monicity. The assertion above can be retained with a proof, or omitted as unnecessary; it should not be described as disproved.

The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U,f)],[I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 (Positive tensor presentation)**.** *Let T_+ and $T_{\mu,+}$ be the sources of Ψ and $\mu\Psi$, respectively. All definitions below use the original ground $\mathbb{S}$. For a* RES *object with underlying set X, put*

$$\Lambda_X = \{\delta \in \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}) : \exists\ \mathbb{S}\text{-}\textit{definable}\ c : X \longrightarrow \delta^\sharp\}.$$